In-match strategies to mitigate the effect of heat on football (soccer) players' health and performance

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THE 2026 FIFA WORLD CUP CHALLENGE

Historical data between 2012 and 2021 reveal the potential weather challenges awaiting the upcoming 2026 Fédération Internationale de Football Association (FIFA) World Cup slated for June-July 2026 in the USA, Canada and Mexico.[1] The Wet Bulb Globe Temperature (WBGT) is projected to range between 30°C and 35°C in six host cities (Atlanta, Dallas, Houston, Kansas City, Miami and Monterrey) in the afternoon (15:00–18:00 hours).[1] To beat the heat, recommended strategies include allowing adequate time to acclimatise, implementing in-match cooling breaks and planning for effective rehydration.[2 3] The most effective method to beat the heat is heat acclimatisation/acclimation.[4] For football players, maintaining internal body temperature above 38.5°C for 60 min with training for a total of 10–14 sessions may ensure adequate heat adaptations.[4] In addition, maintaining euhydration during a football match in the heat is essential and it can reduce internal body temperature and improve physiological function compared with playing in a dehydrated state.[5] Although in-play cooling is advocated,[3] there is a scarcity of literature addressing the most efficient cooling methods,[2] with the majority of available evidence not directly applicable to football. Consequently, the aim of this editorial is to provide a brief discussion of evidence, grounded in realistic studies, pertaining to in-match cooling in football and offer practical solutions for medical and support staff.

WHY IS HOT WEATHER A CHALLENGE IN FOOTBALL?

Playing football elevates core body temperature, and this increase is larger in the heat.[6] Rectal temperatures ranging from 39.5°C to 40.2°C[6] and muscle temperatures reaching as high as 42°C[7] have been reported during football matches played in the heat. Although high speed running may be compromised, peak speed and percentage of successful passes either remain constant or may even improve under these conditions.[6 8] As core temperatures exceed 38.7°C, cognitive function in complex tasks begins to decline.[9] This raises concerns about the potential impact on the quality of technical decision-making in football-related situations, although direct evidence of such an impact is not yet available. Unlike some other sports (ie, endurance running, cycling), football players are able to cool during stoppages in play, the 3 min cooling breaks or at half-time.

HOW TO BEAT THE HEAT IN FOOTBALL MATCHES

To ensure optimal conditions for footballers' well-being and performance, it is advisable for players to keep core temperature below 39°C[10] with skin temperature maintained below 33°C and quadriceps muscle temperature being around 39°C.[10] Only a limited number of studies have successfully replicated exercise protocol and conditions (ie, cooling breaks and half-time) that closely mimic actual football situations. Based on this evidence, the following strategies for cooling seem relevant (box 1):

1. Adjust warm-up duration and structure to promote muscle and cardiovascular adaptations while minimising increases in internal body and skin temperature.[11]
2. Adapt game strategy to attenuate metabolic heat production (eg, reduce high intensity activity and distance covered by players).
3. Use partial-body or whole-body cold water immersion (CWI) to reduce internal body temperature during half-time. It is the most effective method of cooling, resulting in a rapid drop of internal body temperature in the range of 0.25°C–0.35°C per minute.[2]
4. If CWI is impractical, consider the application of ice towels covering a large body surface as the second most effective method for reducing internal body temperature (0.1°C per minute).[2]
5. Use a combination of pre-cooling and mid-cooling techniques, including application of ice towels, head and hand cooling, to reduce internal body temperature and improve second-half sprinting performance.[12]
6. When the reduction in internal body temperature is not feasible due to time restrictions (ie, during 3 min cooling breaks), focus on improving hydration and thermal sensation and reducing skin temperature. To this end, chilled water consumption either alone or in combination with an ice towel applied around the neck[13] and/or head and forearm cooling can help.[14] Using fan and/or skin wetting techniques to reduce skin temperature[15] may also improve the perception of effort and preserve cognitive function.[16]
7. The organiser could consider extending or incorporating more cooling breaks (ie, before extra time) in extreme heat. The organisation representing the professional football players (Fédération Internationale des Associations de Footballeurs Profes-

Box 1 In-match strategies to mitigate the effect of heat on football (soccer) players' health and performance

- Adjust warm-up duration and structure.
- Adapt game strategy.
- Use partial-body or whole-body cold water immersion during half-time.
- Apply ice towels covering a large body surface.
- Use a combination of pre-cooling and mid-cooling techniques.
- During the 3 min cooling breaks, focus on improving hydration and thermal sensation and reducing skin temperature.
- Consider extending or incorporating more cooling breaks in extreme heat.
- Include medical personnel in decision-making.

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sionnels (FIFPRO)) suggests having cooling breaks in the 15th and 60th min of the game to ensure proper hydration.[17] In youth tournaments, a lower threshold (currently at WBGT 32°C) for cooling breaks should be implemented.

8. Recommend the inclusion of medical personnel in decision-making process directly impacting the health and safety of players (eg, match scheduling). Decisions based on healthcare knowledge and the athletes' experiences[18] can prevent the negative impact of heat on players' health.

FOOTBALL PLAY IN THE PERSPECTIVE OF GLOBAL WARMING

As football games are increasingly played in hotter conditions, our recommendations hold relevance beyond the 2026 FIFA World Cup and may apply to players and referees. These recommendations are primarily based on studies involving men, highlighting the necessity for further studies with women and players of all ages. Although the differences in thermoregulation between sex seem to be largely due to differences in anthropometry in hot and humid conditions, women may be at a disadvantage in dry heat, due to lower sweat production.[19] This editorial also underscores the importance of conducting more pragmatic studies. Finally, investigating the challenges posed by additional playing time and multiple extra time matches in FIFA World Cups[20] and other international tournaments played in hot environments is needed.

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